Convex Order and Moment Persuasion

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Introduction

- The rapid growth of literature on Bayesian persuasion
  - Limited progress on multidimensional information design
- Multidimensional moment persuasion with discontinuous objective
  - Sender's objective depends only on posterior means
- Provides the most precise characterization of optimal signal
  - Stronger necessary condition than previous literature

Introduction

- Given two probability measures μ and μ_0 on $\Omega \subset \mathbb{R}^n$ with finite means, μ and μ_0 are in the convex order, denoted by $\mu \preccurlyeq_{cx} \mu_0$, if

$$\int \phi \, d\mu \leq \int \phi \, d\mu_0 \text{ for all convex functions } \phi : \Omega \to \mathbb{R}$$

- μ is mean-preserving contractions of μ_0
- $MPC(\mu_0) := \{\mu \in \Delta(\Omega) \mid \mu \preccurlyeq_{cx} \mu_0\}$

Introduction

- $MPC(\mu_0)$ is also the set of distributions of posterior means
- Moment Persuasion satisfies special mathematical structure
  - a linear functional over the set of mean preserving contractions
- One of the optimum is achieved at an extreme point of $MPC(\mu_0)$

Introduction

This paper studies

- the geometric properties of μ and μ_0 that are in the convex order
- the extreme points of $MPC(\mu_0)$
- the application of extreme points of $MPC(\mu_0)$ into moment persuasion

Product recommendations: the environment

- Platform information:

$$(v, p) \sim U\left([0,1]^2\right).$$

v is the product's value to the consumer, p is the platform's commission from a purchase

- Consumer information:

$$c \sim U[0,1], \qquad c \perp (v, p).$$

The consumer privately knows her purchase cost c

Product recommendations: the environment

- Before the state is realized, the platform commits to a signal based on (v, p), the platform does not observe c
- After signal S, the consumer buys if

$$c \leq \mathbb{E}[v \mid S].$$

A purchase gives the consumer $v - c$ and the platform p

The platform's problem

- Write

$$X_1 = \mathbb{E}[v \mid S], \qquad X_2 = \mathbb{E}[p \mid S].$$

- Independence of c implies

$$\Pr(c \leq X_1 \mid S) = X_1, \qquad \mathbb{E}[p\mathbf{1}_{\{c \leq X_1\}} \mid S] = X_1 X_2.$$

- The platform maximizes a product of posterior means

The platform's problem

- Thus the indirect payoff is $U(x) = x_1 x_2$, and the platform solves

$$\max_{\mu \preccurlyeq_{cx} \mu_0} \int_{[0,1]^2} x_1 x_2 \, d\mu, \qquad \mu_0 = U\big([0,1]^2\big).$$

- Convex order describes the feasible distributions of posterior means

Moment Persuasion(Dworczak and Kolotilin 2024)

- A sender can reveal information about the state to a receiver
- The state of the world ω is distributed according to a prior $\mu_0 \in \Delta(\Omega)$
  - Ω is assumed to be a convex and compact subset of $\mathbb{R}^n$
- The sender chooses a signal π that consists of
  - a signal realization space S
  - a family of distributions $\pi(\cdot \mid \omega)$ over S conditional on the ω

Moment Persuasion

The receiver:

- observes the choice of signal π and the signal realization $s \in S$
- chooses an optimal action that depends on the posterior mean, x

The sender:

- utility is state independent and only depends on receiver's action
- sender's indirect utility $u(x)$ only depends on the posterior mean x
- $u(x)$ is upper semicontinuous and bounded

Strassen's Theorem

- Each signal induces a distribution of posteriors, and hence a distribution of posterior means
- $MPC(\mu_0) := \{\mu \in \Delta(\Omega) \mid \mu \preccurlyeq_{cx} \mu_0\}$
- Strassen's Theorem:
  - For any $\mu \in MPC(\mu_0)$, there exists a signal π that induces μ
  - For any signal π, the distribution of posterior means $\mu \in MPC(\mu_0)$
- $X = S = \Omega$

Moment Persuasion

- Formally, the sender's problem is to choose a distribution over posterior mean beliefs of the receiver, μ, that solves

$$\max_{\mu \in MPC(\mu_0)} \int_X u(x)\, d\mu(x)$$

- One of the optimal solution is achieved at an extreme point of $MPC(\mu_0)$
- Application: firms supply product information to consumers

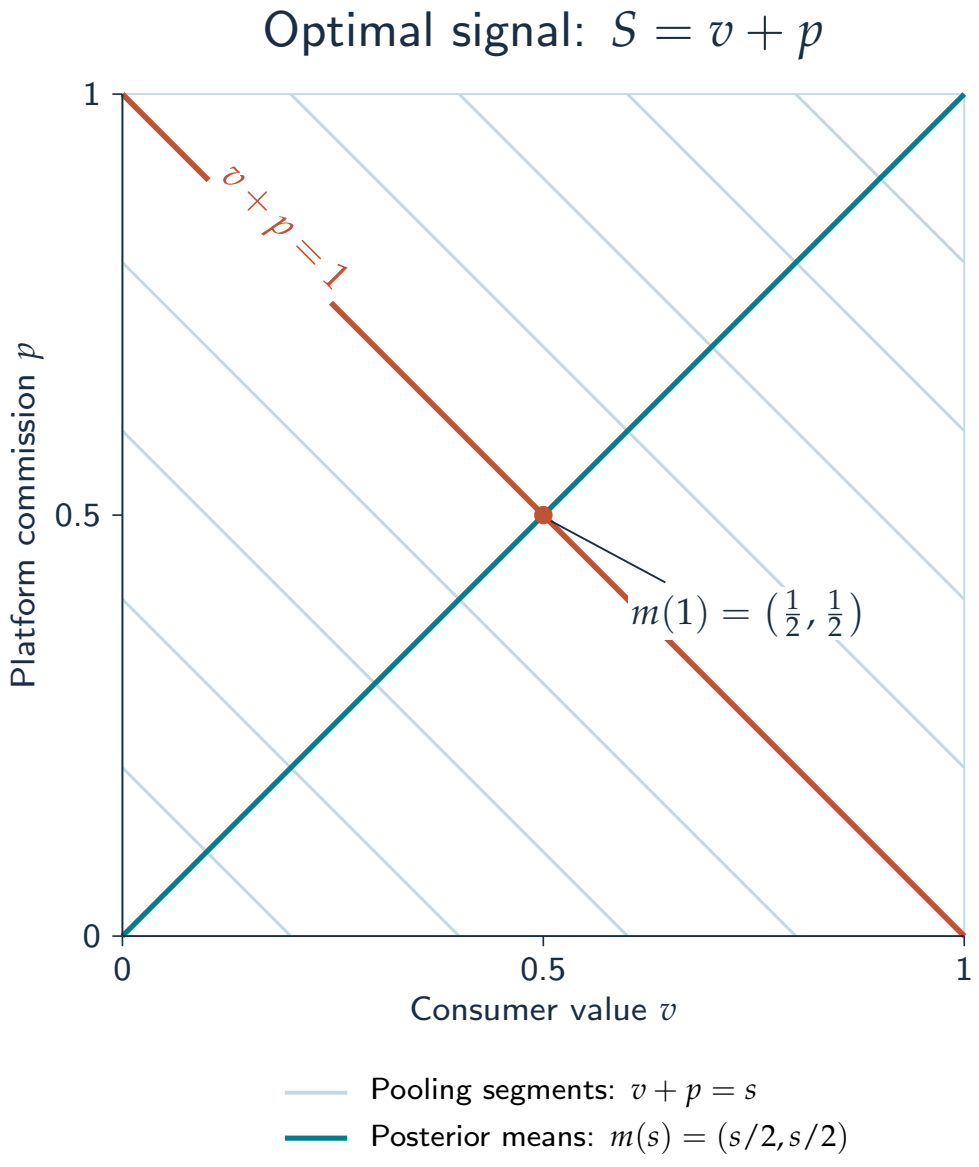
Product recommendations: the optimal signal

- An optimal signal reveals value plus commission: $S = v + p$
- Symmetry gives

$$\mathbb{E}[(v, p) \mid S = s] = \left(\frac{s}{2}, \frac{s}{2}\right).$$

- Each descending segment induces one posterior mean on the diagonal

Product recommendations: the optimal signal



Extreme Points of MPC

Theorem 1: $\mu \in MPC(\mu_0)$ is an extreme point of $MPC(\mu_0)$ only if there exists a collection of disjoint convex and relatively open sets $\{A_x\}_{x \in X}$ such that for μ a.s. x,

- $x \in A_x$
- $\mu_{|A_x}$ is **concentrated** on $cl(A_x)$ concentration
- $\mu_{|A_x}$ has **affinely independent** support affine independence

Note: $\mu_{|A}$ means the conditional distribution of μ on A through the disintegration Theorem

Assumption 1

- **Assumption 1**: $\mathcal{V}$ is μ_0-null set, where $\mathcal{V} := \cup_{\{x \in X : A_x \neq \{x\}\}} rbd(A_x)$.

  Discrete Case

- **Lemma 1**: If $n = 2$ and $\mathcal{V}$ is Lebesgue measurable, then $\mathcal{V}$ is zero Lebesgue measure. If $n \geq 3$, there exists a convex partition such that $\mathcal{V}$ is compact and has positive Lebesgue measure.

Extreme Points of MPC

- **Theorem 2**: Suppose $\mu \in MPC(\mu_0)$ and Assumption 1 holds, then μ is an extreme point of $MPC(\mu_0)$ if and only if there exists a collection of disjoint convex and relatively open sets $\{A_x\}_{x \in X}$ such that for μ-a.s. x,
  - $x \in A_x$
  - $\mu_{|A_x}$ is **concentrated** on $cl(A_x)$
  - $\mu_{|A_x}$ has **affinely independent** support
  - $\mu_{|A_x} \preccurlyeq \mu_{0|A_x}$
  - $\mu_{|A} = \mu_{0|A}$, where $A = \cup_{\{x \in X : A_x = \{x\}\}} A_x$

Definition: Concentration

- We define $\Pi(\mu, \mu_0) = \{p \in \Delta(X \times \Omega) \mid p_X = \mu, p_\Omega = \mu_0\}$
- The subset of martingale transports is

$$\mathcal{M}(\mu, \mu_0) := \{p \in \Pi(\mu, \mu_0) \mid \mathbb{E}[Y \mid Z] = Z \text{ for } (Z, Y) \sim p\}.$$

- For a probability measure μ, $supp(\mu)$ is support of μ
- $\widehat{supp}(\mu) := cl(conv(supp(\mu)))$ the smallest convex closed set that contains support of μ

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Definition: Concentration

- For any given $x \in X$ and $p \in \mathcal{M}(\mu, \mu_0)$, The conditional kernel p_x is defined μ-a.e. by

$$p(dx, d\omega) = \mu(dx) \otimes p_x(d\omega)$$

Given a convex and relatively open set $A \subset X \subset \mathbb{R}^n$, $\mu_{|A}$ is **concentrated** on $cl(A)$ if

- for all $p \in \mathcal{M}(\mu, \mu_0)$ and $\mu_{|A}$-a.s. x, $\widehat{supp}(p_x) \subset cl(A)$
- there exists $\hat{p} \in \mathcal{M}(\mu, \mu_0)$ such that $\widehat{supp}(\hat{p}_x) = cl(A)$ for $\mu_{|A}$-a.s. x.

Convex Order:geometric characterization

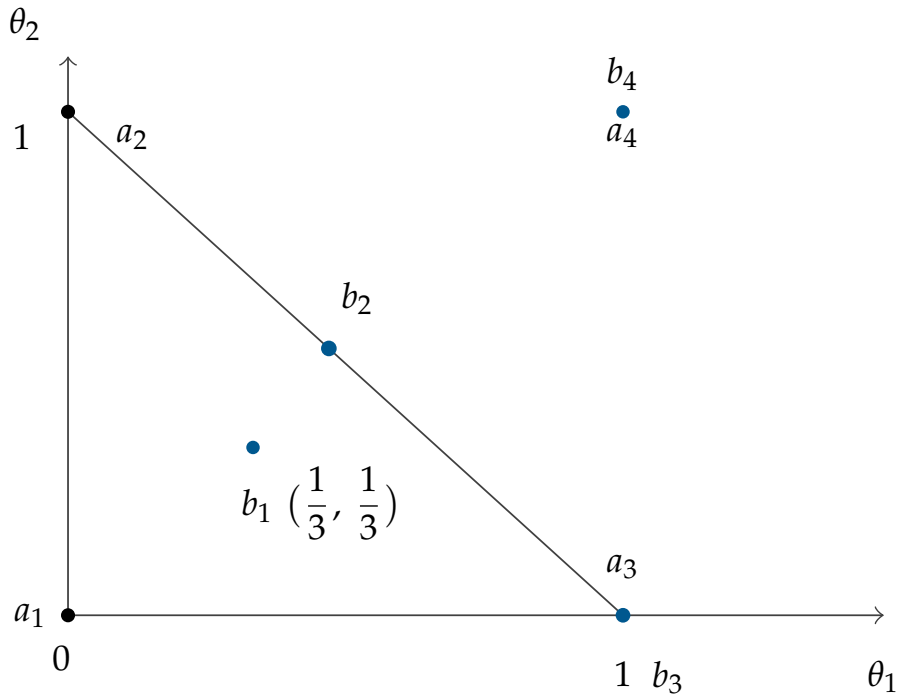
Proposition 1: (March and Touzi 2019) There exists $\hat{p} \in \mathcal{M}(\mu, \mu_0)$ such that for all $p \in \mathcal{M}(\mu, \mu_0)$, $\widehat{supp}(p_x) \subset \widehat{supp}(\hat{p}_x)$ for μ-a.s. x. Furthermore $\widehat{supp}(\hat{p}_x)$ is μ-a.s. unique, and we may choose this kernel so that:

- $x \to \widehat{supp}(\hat{p}_x)$ is analytically measurable $\mathbb{R}^n \to \widehat{\mathcal{K}}$
- $x \in A_x := ri(\widehat{supp}(\hat{p}_x))$ for all $x \in \mathbb{R}^n$, and $\{ri(\widehat{supp}(\hat{p}_x)), x \in \mathbb{R}^n\}$ is a partition of $\mathbb{R}^n$

Corollary 1: $\mu_{\{\mathbb{R}^n \setminus cl(A)\}} = \mu_{0\{\mathbb{R}^n \setminus cl(A)\}}$, where $A := \mathbb{R}^n \setminus \{x \in \mathbb{R}^n : A_x = \{x\}\}$

Discrete Case

- $\mu_0(a_1) = \frac{1}{12},\ \mu_0(a_2) = \frac{5}{24},\ \mu_0(a_3) = \frac{11}{24}, \mu_0(a_4) = \frac{1}{4}.$
- $\mu(b_1) = \frac{1}{4},\ \mu(b_2) = \frac{1}{4},\ \mu(b_3) = \frac{1}{4}, \mu(b_4) = \frac{1}{4}.$
- $A_{b_1} = riconv(a_1, a_2, a_3), A_{b_2} = riconv(a_2, a_3), A_{b_3} = \{b_3\}, A_{b_4} = \{b_4\}.$



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Persuasion Duality

A function $p : X \to R$ is called a price function of μ in the problem (μ_0, u) if the following conditions hold:

- p is convex
- p is L-Lipschitz
- $p(x) \geq u(x)$ for every $x \in X$
- $\mathbb{E}_{x\sim\mu}[p(x)] = \mathbb{E}_{x\sim\mu_0}[p(x)]$
- $supp(\mu) \subset \{x : p(x) = u(x)\}$

Multidimensional Moment Persuasion

We summarize the essential results of Dworczak and Kolotilin (2024) as follows:

- Let (μ_0, u) be a persuasion problem and let $\mu \in MPC(\mu_0)$. If there exists a price function p of μ, then μ is optimal. Furthermore, if u is L-Lipschitz, then there exists an optimal distribution μ for which such a price function exists.

Affine Independence

- We say vectors $x_1, x_2, ..x_r \in R^n$ are affinely independent if $x_2 - x_1, x_3 - x_1, ..x_r - x_1$ are linearly independent
- Alternatively, a finite collection $\{x_1, ..., x_k\} \subset \mathbb{R}^n$ is affinely independent if the unique solution to $\sum_{i=1}^{k} \beta_i x_i = 0$ and $\sum_{i=1}^{k} \beta_i = 0$ is $\beta_1 = \beta_2 = ... = \beta_k = 0$
- A set of affinely independent vectors contains at most $n + 1$ elements

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Extreme Points of MPS

Theorem 3: $\mu \in MPS(\mu_0)$ is an extreme point of $MPS(\mu_0)$ if and only if there exists a collection of disjoint and relatively open simplexes $\{A_x\}_{x \in X}$ such that for μ_0-a.s. x,

- $x \in A_x$
- $\mathcal{M}(\mu_0, \mu)$ is a singleton set,
- for each A_x, p_x is supported on the vertices of A_x for all $x \in A_x$